

Modeling Part I

Setup

```
#install.packages(c("survey", "mgcv", "glmnet", "dplyr"))  
library(survey)
```

Warning: package 'survey' was built under R version 4.5.3

Loading required package: grid

Loading required package: Matrix

Loading required package: survival

Attaching package: 'survey'

The following object is masked from 'package:graphics':

dotchart

```
library(mgcv)
```

Loading required package: nlme

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

```
library(glmnet)
```

Warning: package 'glmnet' was built under R version 4.5.2

Loaded glmnet 4.1-10

```
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.5.2

Attaching package: 'dplyr'

The following object is masked from 'package:nlme':

collapse

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(pROC)
```

Warning: package 'pROC' was built under R version 4.5.3

Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

cov, smooth, var

Survey design

```
df <- readRDS("../data/brfss2024_clean.rds")

# create survey design object
options(survey.lonely.psu = "adjust")

brfss_design <- svydesign(
  id = ~PSU,
  strata = ~STSTR,
  weights = ~WEIGHT,
  data = df,
  nest = TRUE
)
```

Logistic Regression

weighted

```
weighted_glm <- svyglm(
  MEDCOST1 ~ region +
    rurality +
    sex +
    agegroup +
    race +
    income +
    maritalstatus +
    education +
    employment +
    healthstatus +
    physhlth_days +
    menthlth_days +
    depression +
    chronic_count +
    any_disability +
    insurance +
    pers_doc +
    last_checkup,
  design = brfss_design,
```

```
family = quasibinomial()
)

summary(weighted_glm)
```

Call:

```
svyglm(formula = MEDCOST1 ~ region + rurality + sex + agegroup +
  race + income + maritalstatus + education + employment +
  healthstatus + physhlth_days + menthlth_days + depression +
  chronic_count + any_disability + insurance + pers_doc + last_checkup,
  design = brfss_design, family = quasibinomial())
```

Survey design:

```
svydesign(id = ~PSU, strata = ~STSTR, weights = ~WEIGHT, data = df,
  nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-4.691413	0.072929	-64.328	< 2e-16	***
regionNortheast	-0.140600	0.036907	-3.810	0.000139	***
regionMidwest	-0.209880	0.029923	-7.014	2.32e-12	***
regionWest	-0.153937	0.039366	-3.910	9.21e-05	***
ruralityRural	-0.044957	0.044715	-1.005	0.314700	
sexFemale	0.262394	0.028608	9.172	< 2e-16	***
agegroup18-29	0.519326	0.055126	9.421	< 2e-16	***
agegroup30-39	0.604619	0.053207	11.364	< 2e-16	***
agegroup40-49	0.537671	0.053010	10.143	< 2e-16	***
agegroup50-59	0.197687	0.050802	3.891	9.97e-05	***
agegroup70+	-0.739938	0.074944	-9.873	< 2e-16	***
raceBlack_NH	0.301708	0.042209	7.148	8.83e-13	***
raceAsian_NH	-0.009093	0.086366	-0.105	0.916155	
raceAIAN_NH	0.112178	0.121117	0.926	0.354346	
raceHispanic	0.292639	0.039928	7.329	2.32e-13	***
raceOther_NH	0.316576	0.058954	5.370	7.88e-08	***
incomeUpper_mid	0.809395	0.044899	18.027	< 2e-16	***
incomeMiddle	1.075294	0.048742	22.061	< 2e-16	***
incomeLow_mid	1.206664	0.063806	18.911	< 2e-16	***
incomeLow	1.089458	0.069329	15.714	< 2e-16	***
incomeUnknown	0.711937	0.053896	13.209	< 2e-16	***
maritalstatusCurrently_not_married	0.031924	0.031344	1.018	0.308451	
educationSome_college	0.160747	0.035419	4.538	5.67e-06	***

educationHS_grad	0.004378	0.038145	0.115	0.908623	
educationLess_than_HS	0.031385	0.054124	0.580	0.562011	
employmentUnemployed	0.119893	0.050529	2.373	0.017656	*
employmentNot_in_labor	-0.252255	0.051518	-4.896	9.76e-07	***
employmentRetired	-0.649939	0.070911	-9.166	< 2e-16	***
employmentUnable_to_work	-0.384161	0.057048	-6.734	1.65e-11	***
healthstatusVery_good	0.306520	0.051832	5.914	3.35e-09	***
healthstatusGood	0.572824	0.049914	11.476	< 2e-16	***
healthstatusFair	0.940153	0.056894	16.525	< 2e-16	***
healthstatusPoor	0.944822	0.080633	11.718	< 2e-16	***
physlth_days	0.015085	0.001722	8.759	< 2e-16	***
menthlth_days	0.027962	0.001493	18.726	< 2e-16	***
depression	0.292543	0.033016	8.861	< 2e-16	***
chronic_count	0.105422	0.015844	6.654	2.86e-11	***
any_disability	0.522278	0.033828	15.439	< 2e-16	***
insurancePrivate	0.202203	0.051122	3.955	7.64e-05	***
insuranceMedicare	-0.268649	0.054426	-4.936	7.97e-07	***
insuranceMedicaid_CHIP	-0.205516	0.050430	-4.075	4.60e-05	***
insuranceMilitary_VA	-0.808437	0.113346	-7.132	9.88e-13	***
insuranceOther_govt	0.234796	0.058959	3.982	6.83e-05	***
insuranceUninsured	1.366496	0.047731	28.629	< 2e-16	***
pers_docNo	0.238646	0.040784	5.851	4.88e-09	***
last_checkup1_2yrs	0.447835	0.040317	11.108	< 2e-16	***
last_checkup2_5yrs	0.694883	0.050632	13.724	< 2e-16	***
last_checkupOver_5yrs	0.570573	0.055778	10.229	< 2e-16	***
last_checkupNever	0.438859	0.148436	2.957	0.003111	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 0.9516301)

Number of Fisher Scoring iterations: 6

Unweighted

```
unweighted_glm <- glm(
  MEDCOST1 ~ . - PSU - STSTR - WEIGHT,
  data = df,
  family = binomial()
)
```

```
summary(unweighted_glm)
```

Call:

```
glm(formula = MEDCOST1 ~ . - PSU - STSTR - WEIGHT, family = binomial(),  
     data = df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.7545417	0.0372363	-127.686	< 2e-16	***
regionNortheast	-0.1567707	0.0191030	-8.207	2.28e-16	***
regionMidwest	-0.1809101	0.0174242	-10.383	< 2e-16	***
regionWest	-0.0877428	0.0182142	-4.817	1.46e-06	***
ruralityRural	-0.0573046	0.0197335	-2.904	0.00369	**
sexFemale	0.2417438	0.0137339	17.602	< 2e-16	***
agegroup18-29	0.6087968	0.0263054	23.143	< 2e-16	***
agegroup30-39	0.6913887	0.0250124	27.642	< 2e-16	***
agegroup40-49	0.5605220	0.0243452	23.024	< 2e-16	***
agegroup50-59	0.2827411	0.0236364	11.962	< 2e-16	***
agegroup70+	-0.7387136	0.0304268	-24.278	< 2e-16	***
raceBlack_NH	0.3002491	0.0229682	13.072	< 2e-16	***
raceAsian_NH	-0.0041596	0.0431883	-0.096	0.92327	
raceAIAN_NH	0.1809320	0.0448690	4.032	5.52e-05	***
raceHispanic	0.3441004	0.0194568	17.685	< 2e-16	***
raceOther_NH	0.2654812	0.0303501	8.747	< 2e-16	***
incomeUpper_mid	0.7564848	0.0225212	33.590	< 2e-16	***
incomeMiddle	1.1350367	0.0240875	47.121	< 2e-16	***
incomeLow_mid	1.2462585	0.0299033	41.676	< 2e-16	***
incomeLow	1.1149415	0.0337364	33.049	< 2e-16	***
incomeUnknown	0.7335214	0.0273271	26.842	< 2e-16	***
maritalstatusCurrently_not_married	0.0432346	0.0146558	2.950	0.00318	**
educationSome_college	0.0907360	0.0170485	5.322	1.03e-07	***
educationHS_grad	-0.0344860	0.0177934	-1.938	0.05261	.
educationLess_than_HS	0.0425662	0.0262845	1.619	0.10535	
employmentUnemployed	0.1217801	0.0243630	4.999	5.78e-07	***
employmentNot_in_labor	-0.2399319	0.0251680	-9.533	< 2e-16	***
employmentRetired	-0.6805827	0.0291088	-23.381	< 2e-16	***
employmentUnable_to_work	-0.3964941	0.0266539	-14.876	< 2e-16	***
healthstatusVery_good	0.2849701	0.0260830	10.925	< 2e-16	***
healthstatusGood	0.6147484	0.0249571	24.632	< 2e-16	***
healthstatusFair	0.9204390	0.0281932	32.648	< 2e-16	***

healthstatusPoor	0.9865343	0.0369940	26.667	< 2e-16	***
physlth_days	0.0129389	0.0008069	16.036	< 2e-16	***
menthlth_days	0.0270971	0.0007089	38.223	< 2e-16	***
depression	0.2795372	0.0159769	17.496	< 2e-16	***
chronic_count	0.1037303	0.0064330	16.125	< 2e-16	***
any_disability	0.5306721	0.0158186	33.547	< 2e-16	***
insurancePrivate	0.2155210	0.0238903	9.021	< 2e-16	***
insuranceMedicare	-0.2981217	0.0252097	-11.826	< 2e-16	***
insuranceMedicaid_CHIP	-0.3001107	0.0248003	-12.101	< 2e-16	***
insuranceMilitary_VA	-0.8784311	0.0478883	-18.343	< 2e-16	***
insuranceOther_govt	0.1551054	0.0263230	5.892	3.81e-09	***
insuranceUninsured	1.3837834	0.0224937	61.519	< 2e-16	***
pers_docNo	0.2588448	0.0191459	13.520	< 2e-16	***
last_checkup1_2yrs	0.5661592	0.0198051	28.587	< 2e-16	***
last_checkup2_5yrs	0.6945411	0.0244268	28.434	< 2e-16	***
last_checkupOver_5yrs	0.6219309	0.0271060	22.944	< 2e-16	***
last_checkupNever	0.3388418	0.0608578	5.568	2.58e-08	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 229188 on 377794 degrees of freedom
 Residual deviance: 171823 on 377746 degrees of freedom
 AIC: 171921

Number of Fisher Scoring iterations: 6

Comparison

```
# model summary comparison table
weighted_glm_summary <- summary(weighted_glm)$coefficients
unweighted_glm_summary <- summary(unweighted_glm)$coefficients

weighted_glm_df <- data.frame(
  Variable = rownames(weighted_glm_summary),
  Weighted_Estimate = weighted_glm_summary[, 1],
  Weighted_SE = weighted_glm_summary[, 2],
  Weighted_p = weighted_glm_summary[, 4]
)
unweighted_glm_df <- data.frame(
```

```

Variable = rownames(unweighted_glm_summary),
Unweighted_Estimate = unweighted_glm_summary[, 1],
Unweighted_SE = unweighted_glm_summary[, 2],
Unweighted_p = unweighted_glm_summary[, 4]
)

# OR and 95% CI
weighted_glm_df$Weighted_OR <- exp(weighted_glm_df$Weighted_Estimate)
weighted_ci <- exp(confint(weighted_glm))
weighted_glm_df$Weighted_OR_lower <- weighted_ci[, 1]
weighted_glm_df$Weighted_OR_upper <- weighted_ci[, 2]
unweighted_glm_df$Unweighted_OR <- exp(unweighted_glm_df$Unweighted_Estimate)
unweighted_ci <- exp(confint.default(unweighted_glm))
unweighted_glm_df$Unweighted_OR_lower <- unweighted_ci[, 1]
unweighted_glm_df$Unweighted_OR_upper <- unweighted_ci[, 2]

comparison_glm_table <- merge(
  weighted_glm_df,
  unweighted_glm_df,
  by = "Variable"
)

comparison_glm_table <- comparison_glm_table %>%
  mutate(across(where(is.numeric), round, 3))

```

Warning: There was 1 warning in `mutate()`.

i In argument: `across(where(is.numeric), round, 3)`.

Caused by warning:

! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
Supply arguments directly to `fns` through an anonymous function instead.

```

# Previously
across(a:b, mean, na.rm = TRUE)

# Now
across(a:b, \(x) mean(x, na.rm = TRUE))

```

```
comparison_glm_table
```

	Variable	Weighted_Estimate	Weighted_SE	Weighted_p
1	(Intercept)	-4.691	0.073	0.000

2	agegroup18-29	0.519	0.055	0.000
3	agegroup30-39	0.605	0.053	0.000
4	agegroup40-49	0.538	0.053	0.000
5	agegroup50-59	0.198	0.051	0.000
6	agegroup70+	-0.740	0.075	0.000
7	any_disability	0.522	0.034	0.000
8	chronic_count	0.105	0.016	0.000
9	depression	0.293	0.033	0.000
10	educationHS_grad	0.004	0.038	0.909
11	educationLess_than_HS	0.031	0.054	0.562
12	educationSome_college	0.161	0.035	0.000
13	employmentNot_in_labor	-0.252	0.052	0.000
14	employmentRetired	-0.650	0.071	0.000
15	employmentUnable_to_work	-0.384	0.057	0.000
16	employmentUnemployed	0.120	0.051	0.018
17	healthstatusFair	0.940	0.057	0.000
18	healthstatusGood	0.573	0.050	0.000
19	healthstatusPoor	0.945	0.081	0.000
20	healthstatusVery_good	0.307	0.052	0.000
21	incomeLow	1.089	0.069	0.000
22	incomeLow_mid	1.207	0.064	0.000
23	incomeMiddle	1.075	0.049	0.000
24	incomeUnknown	0.712	0.054	0.000
25	incomeUpper_mid	0.809	0.045	0.000
26	insuranceMedicaid_CHIP	-0.206	0.050	0.000
27	insuranceMedicare	-0.269	0.054	0.000
28	insuranceMilitary_VA	-0.808	0.113	0.000
29	insuranceOther_govt	0.235	0.059	0.000
30	insurancePrivate	0.202	0.051	0.000
31	insuranceUninsured	1.366	0.048	0.000
32	last_checkup1_2yrs	0.448	0.040	0.000
33	last_checkup2_5yrs	0.695	0.051	0.000
34	last_checkupNever	0.439	0.148	0.003
35	last_checkupOver_5yrs	0.571	0.056	0.000
36	maritalstatusCurrently_not_married	0.032	0.031	0.308
37	menthlth_days	0.028	0.001	0.000
38	pers_docNo	0.239	0.041	0.000
39	physhlth_days	0.015	0.002	0.000
40	raceAIAN_NH	0.112	0.121	0.354
41	raceAsian_NH	-0.009	0.086	0.916
42	raceBlack_NH	0.302	0.042	0.000
43	raceHispanic	0.293	0.040	0.000
44	raceOther_NH	0.317	0.059	0.000

45	regionMidwest	-0.210	0.030	0.000
46	regionNortheast	-0.141	0.037	0.000
47	regionWest	-0.154	0.039	0.000
48	ruralityRural	-0.045	0.045	0.315
49	sexFemale	0.262	0.029	0.000

	Weighted_OR	Weighted_OR_lower	Weighted_OR_upper	Unweighted_Estimate
1	0.009	0.008	0.011	-4.755
2	1.681	1.509	1.873	0.609
3	1.831	1.649	2.032	0.691
4	1.712	1.543	1.899	0.561
5	1.219	1.103	1.346	0.283
6	0.477	0.412	0.553	-0.739
7	1.686	1.578	1.801	0.531
8	1.111	1.077	1.146	0.104
9	1.340	1.256	1.429	0.280
10	1.004	0.932	1.082	-0.034
11	1.032	0.928	1.147	0.043
12	1.174	1.096	1.259	0.091
13	0.777	0.702	0.860	-0.240
14	0.522	0.454	0.600	-0.681
15	0.681	0.609	0.762	-0.396
16	1.127	1.021	1.245	0.122
17	2.560	2.290	2.862	0.920
18	1.773	1.608	1.956	0.615
19	2.572	2.196	3.013	0.987
20	1.359	1.227	1.504	0.285
21	2.973	2.595	3.405	1.115
22	3.342	2.949	3.788	1.246
23	2.931	2.664	3.225	1.135
24	2.038	1.834	2.265	0.734
25	2.247	2.057	2.453	0.756
26	0.814	0.738	0.899	-0.300
27	0.764	0.687	0.850	-0.298
28	0.446	0.357	0.556	-0.878
29	1.265	1.127	1.420	0.155
30	1.224	1.107	1.353	0.216
31	3.922	3.571	4.306	1.384
32	1.565	1.446	1.694	0.566
33	2.003	1.814	2.212	0.695
34	1.551	1.159	2.075	0.339
35	1.769	1.586	1.974	0.622
36	1.032	0.971	1.098	0.043
37	1.028	1.025	1.031	0.027

38	1.270	1.172	1.375	0.259
39	1.015	1.012	1.019	0.013
40	1.119	0.882	1.418	0.181
41	0.991	0.837	1.174	-0.004
42	1.352	1.245	1.469	0.300
43	1.340	1.239	1.449	0.344
44	1.372	1.223	1.541	0.265
45	0.811	0.765	0.860	-0.181
46	0.869	0.808	0.934	-0.157
47	0.857	0.794	0.926	-0.088
48	0.956	0.876	1.044	-0.057
49	1.300	1.229	1.375	0.242

	Unweighted_SE	Unweighted_p	Unweighted_OR	Unweighted_OR_lower
1	0.037	0.000	0.009	0.008
2	0.026	0.000	1.838	1.746
3	0.025	0.000	1.996	1.901
4	0.024	0.000	1.752	1.670
5	0.024	0.000	1.327	1.267
6	0.030	0.000	0.478	0.450
7	0.016	0.000	1.700	1.648
8	0.006	0.000	1.109	1.095
9	0.016	0.000	1.323	1.282
10	0.018	0.053	0.966	0.933
11	0.026	0.105	1.043	0.991
12	0.017	0.000	1.095	1.059
13	0.025	0.000	0.787	0.749
14	0.029	0.000	0.506	0.478
15	0.027	0.000	0.673	0.638
16	0.024	0.000	1.130	1.077
17	0.028	0.000	2.510	2.375
18	0.025	0.000	1.849	1.761
19	0.037	0.000	2.682	2.494
20	0.026	0.000	1.330	1.263
21	0.034	0.000	3.049	2.854
22	0.030	0.000	3.477	3.279
23	0.024	0.000	3.111	2.968
24	0.027	0.000	2.082	1.974
25	0.023	0.000	2.131	2.039
26	0.025	0.000	0.741	0.706
27	0.025	0.000	0.742	0.706
28	0.048	0.000	0.415	0.378
29	0.026	0.000	1.168	1.109
30	0.024	0.000	1.241	1.184

31	0.022	0.000	3.990	3.818
32	0.020	0.000	1.761	1.694
33	0.024	0.000	2.003	1.909
34	0.061	0.000	1.403	1.246
35	0.027	0.000	1.863	1.766
36	0.015	0.003	1.044	1.015
37	0.001	0.000	1.027	1.026
38	0.019	0.000	1.295	1.248
39	0.001	0.000	1.013	1.011
40	0.045	0.000	1.198	1.097
41	0.043	0.923	0.996	0.915
42	0.023	0.000	1.350	1.291
43	0.019	0.000	1.411	1.358
44	0.030	0.000	1.304	1.229
45	0.017	0.000	0.835	0.806
46	0.019	0.000	0.855	0.823
47	0.018	0.000	0.916	0.884
48	0.020	0.004	0.944	0.908
49	0.014	0.000	1.273	1.240

Unweighted_OR_upper

1	0.009
2	1.935
3	2.097
4	1.837
5	1.390
6	0.507
7	1.754
8	1.123
9	1.365
10	1.000
11	1.099
12	1.132
13	0.826
14	0.536
15	0.709
16	1.185
17	2.653
18	1.942
19	2.884
20	1.399
21	3.258
22	3.687
23	3.262

24	2.197
25	2.227
26	0.778
27	0.780
28	0.456
29	1.230
30	1.300
31	4.170
32	1.831
33	2.101
34	1.581
35	1.964
36	1.075
37	1.029
38	1.345
39	1.015
40	1.308
41	1.084
42	1.412
43	1.466
44	1.384
45	0.864
46	0.888
47	0.949
48	0.982
49	1.308

Logistic regression (with Lasso)

```
x <- model.matrix(
  MEDCOST1 ~ . - PSU - STSTR - WEIGHT,
  data = df
)[, -1]
y <- df$MEDCOST1
```

Weighted

```
weighted_lasso <- cv.glmnet(
  x,
  y,
  family = "binomial",
  alpha = 1,
  nfolds = 10,
  weights = df$WEIGHT
)
```

Unweighted

```
unweighted_lasso <- cv.glmnet(
  x,
  y,
  family = "binomial",
  alpha = 1,
  nfolds = 10
)
```

Comparison

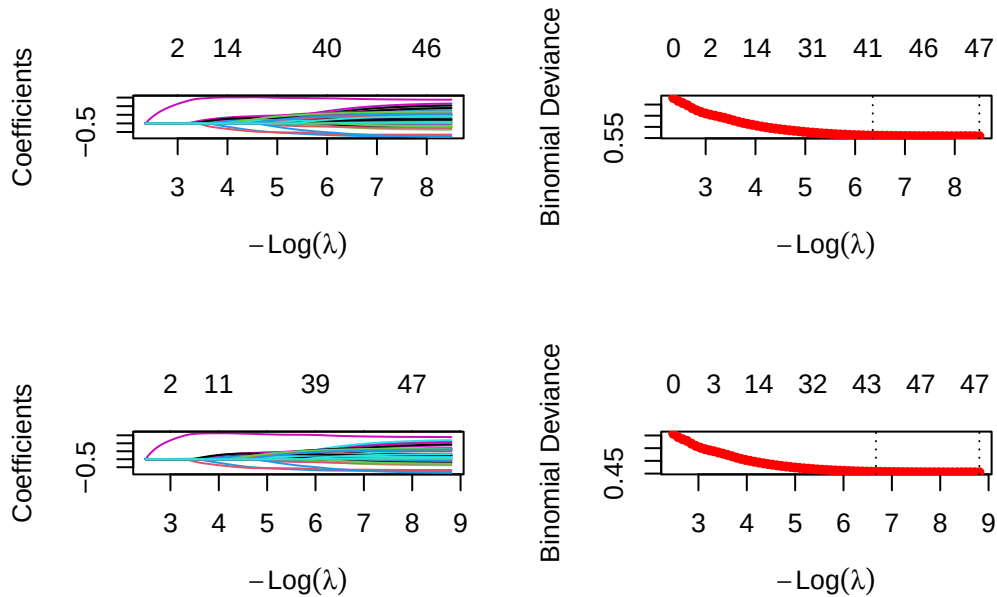
```
# Lasso selected variables
lasso_out <- cbind(coef(weighted_lasso, s = "lambda.1se"),
                  coef(unweighted_lasso, s = "lambda.1se"))
colnames(lasso_out) <- c("Weighted", "Unweighted")
lasso_out
```

49 x 2 sparse Matrix of class "dgCMatrix"

	Weighted	Unweighted
(Intercept)	-3.97171809	-4.16897590
regionNortheast	-0.04520276	-0.05502133
regionMidwest	-0.10647244	-0.07071532
regionWest	-0.05502158	.
ruralityRural	.	.
sexFemale	0.21539382	0.20249238
agegroup18-29	0.26159183	0.41916249
agegroup30-39	0.34408762	0.49676455

agegroup40-49	0.28022991	0.37056071
agegroup50-59	.	0.11313620
agegroup70+	-0.70777306	-0.71666914
raceBlack_NH	0.24506638	0.26734011
raceAsian_NH	.	.
raceAIAN_NH	.	0.06661446
raceHispanic	0.26921170	0.33557313
raceOther_NH	0.21629485	0.19385382
incomeUpper_mid	0.52187928	0.47213996
incomeMiddle	0.76811151	0.81724035
incomeLow_mid	0.84236399	0.87381321
incomeLow	0.70928097	0.72044214
incomeUnknown	0.36064515	0.37838262
maritalstatusCurrently_not_married	0.06339611	0.07244849
educationSome_college	0.11466275	0.07566855
educationHS_grad	.	.
educationLess_than_HS	.	0.03803523
employmentUnemployed	0.14344190	0.15258427
employmentNot_in_labor	-0.12155171	-0.11218073
employmentRetired	-0.63196653	-0.65253400
employmentUnable_to_work	-0.22724191	-0.24794276
healthstatusVery_good	.	.
healthstatusGood	0.28567062	0.33901454
healthstatusFair	0.64989768	0.63912565
healthstatusPoor	0.61184944	0.66942191
physhlth_days	0.01369916	0.01201450
menthlth_days	0.02922259	0.02830326
depression	0.28403601	0.27533954
chronic_count	0.06424655	0.07425054
any_disability	0.50163378	0.50753869
insurancePrivate	0.15684462	0.17860458
insuranceMedicare	-0.20498807	-0.23975833
insuranceMedicaid_CHIP	-0.03329055	-0.13657095
insuranceMilitary_VA	-0.59012116	-0.66696964
insuranceOther_govt	0.26548042	0.19285158
insuranceUninsured	1.43630336	1.45051926
pers_docNo	0.26174271	0.27670109
last_checkup1_2yrs	0.37580298	0.49773805
last_checkup2_5yrs	0.60860407	0.61766945
last_checkupOver_5yrs	0.46633387	0.52737243
last_checkupNever	0.27399097	0.20299015

```
# Lasso plot
par(mfrow=c(2,2))
plot(weighted_lasso$glmnet.fit, xvar = "lambda")
plot(weighted_lasso)
plot(unweighted_lasso$glmnet.fit, xvar = "lambda")
plot(unweighted_lasso)
```



```
# AUC comparison
weighted_lasso_pred <- predict(weighted_lasso, newx = x, s = "lambda.1se", type = "response")
unweighted_lasso_pred <- predict(unweighted_lasso, newx = x, s = "lambda.1se", type = "response")
weighted_lasso_roc_obj <- roc(y, as.numeric(weighted_lasso_pred))
```

Setting levels: control = 0, case = 1

Setting direction: controls < cases

```
unweighted_lasso_roc_obj <- roc(y, as.numeric(unweighted_lasso_pred))
```

Setting level: control = 0, case = 1

Setting direction: controls < cases


```
auc(weighted_lasso_roc_obj)
```

Area under the curve: 0.847

```
auc(unweighted_lasso_roc_obj)
```

Area under the curve: 0.8479

GAM

```
df$survey_weight <- weights(brfss_design)
```

Weighted

```
weighted_gam <- gam(  
  MEDCOST1 ~  
    s(physhlth_days) +  
    s(menthlth_days) +  
    region +  
    rurality +  
    sex +  
    agegroup +  
    race +  
    income +  
    maritalstatus +  
    education +  
    employment +  
    healthstatus +  
    depression +  
    chronic_count +  
    any_disability +  
    insurance +  
    pers_doc +  
    last_checkup,  
  data = df,  
  family = binomial(),
```

```
weights = survey_weight
)
```

```
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```
summary(weighted_gam)
```

```
Family: binomial
Link function: logit
```

```
Formula:
```

```
MEDCOST1 ~ s(physhlth_days) + s(menthlth_days) + region + rurality +
sex + agegroup + race + income + maritalstatus + education +
employment + healthstatus + depression + chronic_count +
any_disability + insurance + pers_doc + last_checkup
```

```
Parametric coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.4252490	0.0014339	-3086.158	< 2e-16	***
regionNortheast	-0.1464745	0.0007432	-197.087	< 2e-16	***
regionMidwest	-0.2102990	0.0006663	-315.612	< 2e-16	***
regionWest	-0.1618492	0.0006349	-254.910	< 2e-16	***
ruralityRural	-0.0357669	0.0010160	-35.202	< 2e-16	***
sexFemale	0.2368639	0.0005150	459.952	< 2e-16	***
agegroup18-29	0.4289305	0.0010344	414.655	< 2e-16	***
agegroup30-39	0.5527645	0.0010012	552.090	< 2e-16	***
agegroup40-49	0.5070943	0.0009913	511.537	< 2e-16	***
agegroup50-59	0.1875506	0.0009912	189.221	< 2e-16	***
agegroup70+	-0.7250500	0.0014226	-509.665	< 2e-16	***
raceBlack_NH	0.3084706	0.0007697	400.742	< 2e-16	***
raceAsian_NH	0.0057069	0.0012196	4.679	2.88e-06	***
raceAIAN_NH	0.1290606	0.0020180	63.956	< 2e-16	***
raceHispanic	0.3099825	0.0006592	470.216	< 2e-16	***
raceOther_NH	0.3293949	0.0011116	296.332	< 2e-16	***
incomeUpper_mid	0.8121216	0.0008347	972.943	< 2e-16	***

incomeMiddle	1.0867506	0.0008897	1221.506	< 2e-16	***
incomeLow_mid	1.2257233	0.0011022	1112.059	< 2e-16	***
incomeLow	1.1101114	0.0012364	897.856	< 2e-16	***
incomeUnknown	0.7409922	0.0009879	750.035	< 2e-16	***
maritalstatusCurrently_not_married	0.0150710	0.0005529	27.258	< 2e-16	***
educationSome_college	0.1780219	0.0006900	257.993	< 2e-16	***
educationHS_grad	0.0445198	0.0007297	61.010	< 2e-16	***
educationLess_than_HS	0.0796465	0.0009198	86.591	< 2e-16	***
employmentUnemployed	0.1147975	0.0008605	133.404	< 2e-16	***
employmentNot_in_labor	-0.2687185	0.0008345	-322.005	< 2e-16	***
employmentRetired	-0.6296229	0.0012759	-493.479	< 2e-16	***
employmentUnable_to_work	-0.3762555	0.0010183	-369.479	< 2e-16	***
healthstatusVery_good	0.2400555	0.0009380	255.911	< 2e-16	***
healthstatusGood	0.4762628	0.0008938	532.831	< 2e-16	***
healthstatusFair	0.7989683	0.0010101	790.950	< 2e-16	***
healthstatusPoor	0.9076656	0.0013613	666.783	< 2e-16	***
depression	0.2204825	0.0006102	361.316	< 2e-16	***
chronic_count	0.0960478	0.0002610	368.035	< 2e-16	***
any_disability	0.4880704	0.0005849	834.410	< 2e-16	***
insurancePrivate	0.2015174	0.0008693	231.816	< 2e-16	***
insuranceMedicare	-0.2544491	0.0010016	-254.049	< 2e-16	***
insuranceMedicaid_CHIP	-0.1875772	0.0009283	-202.075	< 2e-16	***
insuranceMilitary_VA	-0.7877300	0.0016808	-468.656	< 2e-16	***
insuranceOther_govt	0.2455465	0.0009365	262.187	< 2e-16	***
insuranceUninsured	1.3936276	0.0008082	1724.397	< 2e-16	***
pers_docNo	0.2623007	0.0006743	388.985	< 2e-16	***
last_checkup1_2yrs	0.4485660	0.0007295	614.925	< 2e-16	***
last_checkup2_5yrs	0.6912130	0.0008651	799.008	< 2e-16	***
last_checkupOver_5yrs	0.5748146	0.0009925	579.147	< 2e-16	***
last_checkupNever	0.4798280	0.0020827	230.389	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(physhlth_days)	8.771	8.978	630784	<2e-16 ***
s(menthlth_days)	8.971	9.000	1329525	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.218 Deviance explained = 24.2%

UBRE = 312.1 Scale est. = 1 n = 377795

Unweighted

```
unweighted_gam <- gam(
  MEDCOST1 ~
    s(physhlth_days) +
    s(menthlth_days) +
    region +
    rurality +
    sex +
    agegroup +
    race +
    income +
    maritalstatus +
    education +
    employment +
    healthstatus +
    depression +
    chronic_count +
    any_disability +
    insurance +
    pers_doc +
    last_checkup,
  data = df,
  family = binomial()
)

summary(unweighted_gam)
```

Family: binomial

Link function: logit

Formula:

```
MEDCOST1 ~ s(physhlth_days) + s(menthlth_days) + region + rurality +
  sex + agegroup + race + income + maritalstatus + education +
  employment + healthstatus + depression + chronic_count +
  any_disability + insurance + pers_doc + last_checkup
```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.479294	0.037651	-118.969	< 2e-16 ***

regionNortheast	-0.164302	0.019149	-8.580	< 2e-16	***
regionMidwest	-0.180058	0.017471	-10.306	< 2e-16	***
regionWest	-0.100298	0.018264	-5.492	3.98e-08	***
ruralityRural	-0.046249	0.019778	-2.338	0.019365	*
sexFemale	0.210297	0.013793	15.246	< 2e-16	***
agegroup18-29	0.495991	0.026624	18.629	< 2e-16	***
agegroup30-39	0.616120	0.025199	24.450	< 2e-16	***
agegroup40-49	0.515569	0.024455	21.082	< 2e-16	***
agegroup50-59	0.260808	0.023703	11.003	< 2e-16	***
agegroup70+	-0.716612	0.030475	-23.515	< 2e-16	***
raceBlack_NH	0.308330	0.023028	13.389	< 2e-16	***
raceAsian_NH	0.023785	0.043358	0.549	0.583301	
raceAIAN_NH	0.190704	0.044954	4.242	2.21e-05	***
raceHispanic	0.373405	0.019518	19.131	< 2e-16	***
raceOther_NH	0.277115	0.030374	9.123	< 2e-16	***
incomeUpper_mid	0.754147	0.022567	33.418	< 2e-16	***
incomeMiddle	1.139766	0.024145	47.205	< 2e-16	***
incomeLow_mid	1.252882	0.029966	41.810	< 2e-16	***
incomeLow	1.129480	0.033780	33.436	< 2e-16	***
incomeUnknown	0.759907	0.027397	27.737	< 2e-16	***
maritalstatusCurrently_not_married	0.029592	0.014710	2.012	0.044256	*
educationSome_college	0.110348	0.017105	6.451	1.11e-10	***
educationHS_grad	0.004481	0.017890	0.250	0.802225	
educationLess_than_HS	0.093740	0.026436	3.546	0.000391	***
employmentUnemployed	0.115549	0.024432	4.729	2.25e-06	***
employmentNot_in_labor	-0.252161	0.025221	-9.998	< 2e-16	***
employmentRetired	-0.667939	0.029183	-22.888	< 2e-16	***
employmentUnable_to_work	-0.380368	0.026614	-14.292	< 2e-16	***
healthstatusVery_good	0.212824	0.026284	8.097	5.63e-16	***
healthstatusGood	0.504394	0.025280	19.952	< 2e-16	***
healthstatusFair	0.768383	0.028628	26.840	< 2e-16	***
healthstatusPoor	0.937450	0.037009	25.330	< 2e-16	***
depression	0.201984	0.016057	12.579	< 2e-16	***
chronic_count	0.092328	0.006447	14.322	< 2e-16	***
any_disability	0.497305	0.015845	31.385	< 2e-16	***
insurancePrivate	0.220508	0.023964	9.201	< 2e-16	***
insuranceMedicare	-0.283291	0.025266	-11.212	< 2e-16	***
insuranceMedicaid_CHIP	-0.284149	0.024809	-11.453	< 2e-16	***
insuranceMilitary_VA	-0.853746	0.047801	-17.860	< 2e-16	***
insuranceOther_govt	0.165974	0.026374	6.293	3.11e-10	***
insuranceUninsured	1.420634	0.022668	62.672	< 2e-16	***
pers_docNo	0.280382	0.019228	14.582	< 2e-16	***
last_checkup1_2yrs	0.563576	0.019865	28.370	< 2e-16	***

last_checkup2_5yrs	0.689311	0.024527	28.105	< 2e-16 ***
last_checkupOver_5yrs	0.627548	0.027239	23.039	< 2e-16 ***
last_checkupNever	0.379766	0.061315	6.194	5.88e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(physhlth_days)	6.773	7.622	758.7	<2e-16 ***
s(menthlth_days)	8.311	8.815	1875.8	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.21 Deviance explained = 25.6%
 UBRE = -0.54815 Scale est. = 1 n = 377795

Comparison

```
# coefficients comparison table
weighted_gam_coef <- summary(weighted_gam)$p.coef
unweighted_gam_coef <- summary(unweighted_gam)$p.coef

gam_comparison <- data.frame(
  Variable = names(weighted_gam_coef),
  Weighted_Est = weighted_gam_coef,
  Unweighted_Est = unweighted_gam_coef,
  Weighted_OR = exp(weighted_gam_coef),
  Unweighted_OR = exp(unweighted_gam_coef)
)
gam_comparison
```

	Variable
(Intercept)	(Intercept)
regionNortheast	regionNortheast
regionMidwest	regionMidwest
regionWest	regionWest
ruralityRural	ruralityRural
sexFemale	sexFemale
agegroup18-29	agegroup18-29
agegroup30-39	agegroup30-39

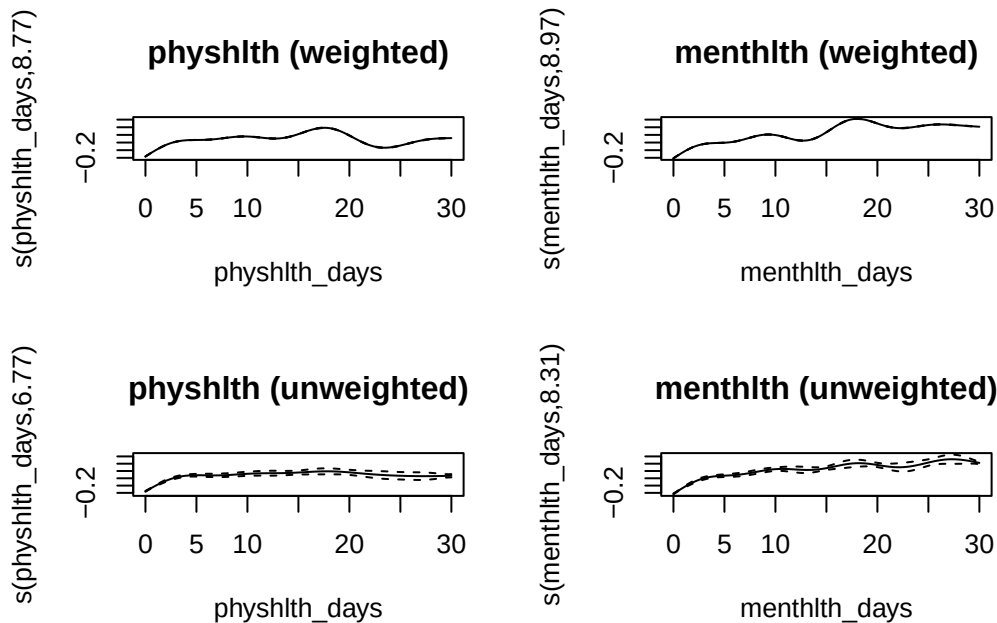
agegroup40-49		agegroup40-49	
agegroup50-59		agegroup50-59	
agegroup70+		agegroup70+	
raceBlack_NH		raceBlack_NH	
raceAsian_NH		raceAsian_NH	
raceAIAN_NH		raceAIAN_NH	
raceHispanic		raceHispanic	
raceOther_NH		raceOther_NH	
incomeUpper_mid		incomeUpper_mid	
incomeMiddle		incomeMiddle	
incomeLow_mid		incomeLow_mid	
incomeLow		incomeLow	
incomeUnknown		incomeUnknown	
maritalstatusCurrently_not_married	maritalstatusCurrently_not_married	maritalstatusCurrently_not_married	
educationSome_college		educationSome_college	
educationHS_grad		educationHS_grad	
educationLess_than_HS		educationLess_than_HS	
employmentUnemployed		employmentUnemployed	
employmentNot_in_labor		employmentNot_in_labor	
employmentRetired		employmentRetired	
employmentUnable_to_work		employmentUnable_to_work	
healthstatusVery_good		healthstatusVery_good	
healthstatusGood		healthstatusGood	
healthstatusFair		healthstatusFair	
healthstatusPoor		healthstatusPoor	
depression		depression	
chronic_count		chronic_count	
any_disability		any_disability	
insurancePrivate		insurancePrivate	
insuranceMedicare		insuranceMedicare	
insuranceMedicaid_CHIP		insuranceMedicaid_CHIP	
insuranceMilitary_VA		insuranceMilitary_VA	
insuranceOther_govt		insuranceOther_govt	
insuranceUninsured		insuranceUninsured	
pers_docNo		pers_docNo	
last_checkup1_2yrs		last_checkup1_2yrs	
last_checkup2_5yrs		last_checkup2_5yrs	
last_checkupOver_5yrs		last_checkupOver_5yrs	
last_checkupNever		last_checkupNever	
	Weighted_Est	Unweighted_Est	Weighted_OR
(Intercept)	-4.425248951	-4.479294206	0.01197123
regionNortheast	-0.146474470	-0.164302099	0.86374778
regionMidwest	-0.210299040	-0.180058334	0.81034189

regionWest	-0.161849247	-0.100297794	0.85056942
ruralityRural	-0.035766916	-0.046249384	0.96486516
sexFemale	0.236863850	0.210297303	1.26726857
agegroup18-29	0.428930522	0.495991265	1.53561434
agegroup30-39	0.552764531	0.616119509	1.73805128
agegroup40-49	0.507094331	0.515568931	1.66045943
agegroup50-59	0.187550601	0.260807546	1.20629129
agegroup70+	-0.725049951	-0.716612416	0.48430038
raceBlack_NH	0.308470640	0.308329858	1.36134154
raceAsian_NH	0.005706876	0.023784716	1.00572319
raceAIAN_NH	0.129060582	0.190703947	1.13775905
raceHispanic	0.309982513	0.373404602	1.36340127
raceOther_NH	0.329394919	0.277114985	1.39012673
incomeUpper_mid	0.812121604	0.754146611	2.25268222
incomeMiddle	1.086750550	1.139766294	2.96462500
incomeLow_mid	1.225723313	1.252881740	3.40662925
incomeLow	1.110111386	1.129480276	3.03469640
incomeUnknown	0.740992172	0.759906910	2.09801607
maritalstatusCurrently_not_married	0.015071038	0.029592298	1.01518518
educationSome_college	0.178021877	0.110347546	1.19485146
educationHS_grad	0.044519752	0.004481003	1.04552563
educationLess_than_HS	0.079646455	0.093740210	1.08290414
employmentUnemployed	0.114797520	0.115548691	1.12164630
employmentNot_in_labor	-0.268718494	-0.252160935	0.76435840
employmentRetired	-0.629622902	-0.667938660	0.53279268
employmentUnable_to_work	-0.376255491	-0.380368350	0.68642693
healthstatusVery_good	0.240055544	0.212824110	1.27131976
healthstatusGood	0.476262802	0.504393555	1.61004608
healthstatusFair	0.798968329	0.768382580	2.22324609
healthstatusPoor	0.907665583	0.937450198	2.47852985
depression	0.220482482	0.201983536	1.24667809
chronic_count	0.096047773	0.092327649	1.10081165
any_disability	0.488070385	0.497305032	1.62916952
insurancePrivate	0.201517367	0.220508189	1.22325748
insuranceMedicare	-0.254449135	-0.283290647	0.77534349
insuranceMedicaid_CHIP	-0.187577218	-0.284149230	0.82896510
insuranceMilitary_VA	-0.787730034	-0.853745843	0.45487618
insuranceOther_govt	0.245546499	0.165973947	1.27831972
insuranceUninsured	1.393627584	1.420633597	4.02944071
pers_docNo	0.262300660	0.280382155	1.29991732
last_checkup1_2yrs	0.448565984	0.563575597	1.56606481
last_checkup2_5yrs	0.691213010	0.689310741	1.99613540
last_checkupOver_5yrs	0.574814598	0.627547870	1.77680107

last_checkupNever	0.479828036	0.379765601	1.61579652
	Unweighted_OR		
(Intercept)	0.01134142		
regionNortheast	0.84848566		
regionMidwest	0.83522149		
regionWest	0.90456800		
ruralityRural	0.95480382		
sexFemale	1.23404489		
agegroup18-29	1.64212521		
agegroup30-39	1.85172847		
agegroup40-49	1.67459096		
agegroup50-59	1.29797784		
agegroup70+	0.48840397		
raceBlack_NH	1.36114990		
raceAsian_NH	1.02406983		
raceAIAN_NH	1.21010114		
raceHispanic	1.45267198		
raceOther_NH	1.31931806		
incomeUpper_mid	2.12579662		
incomeMiddle	3.12603771		
incomeLow_mid	3.50041573		
incomeLow	3.09404803		
incomeUnknown	2.13807718		
maritalstatusCurrently_not_married	1.03003450		
educationSome_college	1.11666610		
educationHS_grad	1.00449106		
educationLess_than_HS	1.09827439		
employmentUnemployed	1.12248917		
employmentNot_in_labor	0.77711966		
employmentRetired	0.51276447		
employmentUnable_to_work	0.68360956		
healthstatusVery_good	1.23716703		
healthstatusGood	1.65598096		
healthstatusFair	2.15627583		
healthstatusPoor	2.55346229		
depression	1.22382786		
chronic_count	1.09672410		
any_disability	1.64428400		
insurancePrivate	1.24671013		
insuranceMedicare	0.75330081		
insuranceMedicaid_CHIP	0.75265432		
insuranceMilitary_VA	0.42581690		
insuranceOther_govt	1.18054234		

insuranceUninsured	4.13974254
pers_docNo	1.32363555
last_checkup1_2yrs	1.75694340
last_checkup2_5yrs	1.99234182
last_checkupOver_5yrs	1.87301207
last_checkupNever	1.46194187

```
# smooth term plot
par(mfrow = c(2, 2))
plot(weighted_gam, select = 1, main = "physhlth (weighted)")
plot(weighted_gam, select = 2, main = "menthlth (weighted)")
plot(unweighted_gam, select = 1, main = "physhlth (unweighted)")
plot(unweighted_gam, select = 2, main = "menthlth (unweighted)")
```



```
# AUC comparison
weighted_gam_pred <- predict(weighted_gam, type = "response")
unweighted_gam_pred <- predict(unweighted_gam, type = "response")
weighted_gam_roc_obj <- roc(y, as.numeric(weighted_gam_pred))
```

Setting levels: control = 0, case = 1

Setting direction: controls < cases

```
unweighted_gam_roc_obj <- roc(y, as.numeric(unweighted_gam_pred))
```

```
Setting levels: control = 0, case = 1  
Setting direction: controls < cases
```

```
auc(weighted_gam_roc_obj)
```

```
Area under the curve: 0.8501
```

```
auc(unweighted_gam_roc_obj)
```

```
Area under the curve: 0.851
```

```
# p-values for smooth terms  
summary(weighted_gam)$s.pv
```

```
[1] 0 0
```

```
summary(unweighted_gam)$s.pv
```

```
[1] 0 0
```

Final model comparison

```
get_auc <- function(y, pred_prob) {  
  roc_obj <- roc(y, as.numeric(pred_prob))  
  auc(roc_obj)  
}  
  
pred_w_glm <- predict(weighted_glm, type = "response")  
auc_w_glm <- get_auc(y, pred_w_glm)
```

```
Setting levels: control = 0, case = 1
```

```
Setting direction: controls < cases
```

```
pred_u_glm <- predict(unweighted_glm, type = "response")
auc_u_glm <- get_auc(y, pred_u_glm)
```

Setting levels: control = 0, case = 1
Setting direction: controls < cases

```
pred_w_lasso <- predict(weighted_lasso, newx = x, s = "lambda.1se", type = "response")
auc_w_lasso <- get_auc(y, pred_w_lasso)
```

Setting levels: control = 0, case = 1
Setting direction: controls < cases

```
pred_u_lasso <- predict(unweighted_lasso, newx = x, s = "lambda.1se", type = "response")
auc_u_lasso <- get_auc(y, pred_u_lasso)
```

Setting levels: control = 0, case = 1
Setting direction: controls < cases

```
pred_w_gam <- predict(weighted_gam, type = "response")
auc_w_gam <- get_auc(y, pred_w_gam)
```

Setting levels: control = 0, case = 1
Setting direction: controls < cases

```
pred_u_gam <- predict(unweighted_gam, type = "response")
auc_u_gam <- get_auc(y, pred_u_gam)
```

Setting levels: control = 0, case = 1
Setting direction: controls < cases

```
model_comparison <- data.frame(
  Model = c(
    "Logistic (Weighted)",
    "Logistic (Unweighted)",
    "LASSO (Weighted)",
    "LASSO (Unweighted)",
    "GAM (Weighted)",
    "GAM (Unweighted)"
  )
)
```

```

),
AUC = c(
  auc_w_glm,
  auc_u_glm,
  auc_w_lasso,
  auc_u_lasso,
  auc_w_gam,
  auc_u_gam
)
)

model_comparison

```

	Model	AUC
1	Logistic (Weighted)	0.8479442
2	Logistic (Unweighted)	0.8486653
3	LASSO (Weighted)	0.8469877
4	LASSO (Unweighted)	0.8479282
5	GAM (Weighted)	0.8501023
6	GAM (Unweighted)	0.8509874